

# Investigating a NetSupport Manager RAT Infection Using Zeek Logs (Evidence-Based Analysis)

## Introduction

Remote Access Trojans (RATs) such as NetSupport Manager are often abused by threat actors to gain persistent control over compromised systems. Detecting such activity within a network requires precise, evidence-driven analysis—especially when working with limited telemetry.

This article presents a structured investigation of suspected NetSupport Manager RAT activity using Zeek network logs, with a strict focus on evidence-based findings. No external enrichment, assumptions, or speculative correlations are used—only what is directly observable in the logs.

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## Scenario Overview

The investigation begins with a known malicious network indicator:

- RAT IP Address: 45.131.214[.]85
- Port: 443
- Timestamp: February 28, 2026, 19:55 UTC

## Environment Details

- LAN Segment: 10.2.28.0/24
  - Gateway: 10.2.28.1
  - Broadcast: 10.2.28.255
  - Domain: easyas123.tech
  - Active Directory Domain: EASYAS123
  - Domain Controller: 10.2.28.2 (EASYAS123-DC)
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## Investigation Methodology

The analysis is performed using the following Zeek logs:

- conn.log → Network communication metadata
- dhcp.log → Host identification (MAC address, hostname)
- kerberos.log → Authentication activity

Each conclusion is derived strictly from these logs.

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### Identifying the Infected Host IP

To determine which internal system communicated with the known malicious IP, we analyze the conn.log.

Command: `cat conn.log | zeek-cut id.orig_h id.resp_h | grep -E '45.131.214.85'`

```
grep -E '45.131.214.85'
10.2.28.88      45.131.214.85
```

Infected Client IP: **10.2.28[.]88**

This confirms that the internal host 10.2.28.88 initiated or participated in communication with the RAT infrastructure.

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### Determining the MAC Address

Next, we correlate the identified IP address with DHCP logs to obtain the physical identifier.

Command: `cat dhcp.log | zeek-cut mac client_addr`

```
00:e0:4c:68:08:00      -
00:19:d1:b2:4d:ad      10.2.28.88
```

MAC Address: **00:19:d1:b2:4d:ad**

This uniquely identifies the network interface associated with the compromised system.

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## Identifying the Hostname

The hostname of the system is also available in the DHCP logs.

Command: `cat dhcp.log | zeek-cut host_name client_addr`

```
brads-MBP      -  
DESKTOP-TEYQ2NR 10.2.28.88
```

Hostname: **DESKTOP-TEYQ2NR**

This suggests the compromised asset is a Windows-based endpoint.

## Combined Extraction (Efficient Approach)

Both MAC address and hostname can be extracted together:

Command: `cat dhcp.log | zeek-cut mac host_name client_addr`

```
00:e0:4c:68:08:00 brads-MBP      -  
00:19:d1:b2:4d:ad  DESKTOP-TEYQ2NR 10.2.28.88
```

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## Identifying the User Account (Username Only)

To determine which user account was active on the system, we analyze Kerberos authentication logs.

Command: `cat kerberos.log | zeek-cut client | sort | uniq`

```
-  
broLf/EASYAS123  
broLf/EASYAS123.TECH
```

## Explanation

- The client field represents the principal initiating authentication.
- `sort` and `uniq` remove duplicate entries.
- Values typically appear in the format:  
username/domain@REALM

Username: **broLf**

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**Important Note on User Identity**

**What Can Be Determined**

- Username (brolf)
- Authentication activity patterns

**What Cannot Be Determined**

Zeek logs do not inherently provide full user identity details, such as:

- Full name (e.g., *Becka Rolf*)
- Display name
- Directory attributes (e.g., LDAP/AD metadata)

Such information is only available if explicitly captured within application-layer logs (e.g., HTTP, FTP). It can be found in deep-inspection tools like Wireshark.

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**Summary of Evidence-Based Findings**

| Attribute   | Value             |
|-------------|-------------------|
| Infected IP | 10.2.28.88        |
| MAC Address | 00:19:d1:b2:4d:ad |
| Hostname    | DESKTOP-TEYQ2NR   |
| Username    | brolf             |

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**Key Takeaways**

- Zeek logs provide strong network-level visibility, enabling identification of compromised hosts without endpoint telemetry.
  - Correlation across logs (conn, DHCP, Kerberos) is critical for building a complete picture.
  - Evidence-based investigation prevents false assumptions, ensuring accuracy in incident response.
  - User attribution is limited without application-layer or directory service logs.
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## **Conclusion**

This investigation demonstrates how a suspected NetSupport Manager RAT infection can be analyzed using only Zeek logs. By systematically correlating network communication, DHCP assignments, and authentication data, we identified the compromised host and associated user account with high confidence—without relying on external enrichment.

This approach is especially valuable in constrained environments where only network telemetry is available, reinforcing the importance of structured, log-driven threat hunting.

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